

Question number	Answer	Mark
4(a)(i)	The only correct answer is C <i>A is incorrect because light-dependent reaction takes place on the thylakoid membranes which is T and the light-independent reactions in the stroma which is R</i> <i>B is incorrect because light-dependent reaction takes place on the thylakoid membranes which is T and the light-independent reactions in the stroma which is R</i> <i>D is incorrect because light-dependent reaction takes place on the thylakoid membranes which is T and the light-independent reactions in the stroma which is R</i>	(1)

Question number	Answer	Mark
4(a)(ii)	The only correct answer is C <i>A is incorrect because hydrogen ions accumulate in the thylakoid space which is S</i> <i>B is incorrect because hydrogen ions accumulate in the thylakoid space which is S</i> <i>D is incorrect because hydrogen ions accumulate in the thylakoid space which is S</i>	(1)

Question number	Answer	Additional guidance	Mark
4(a)(iii)	An answer that includes the following points: - chloroplast measured as 9.6cm and converted to 96 000 μm (1) - value divided by 16 000 (1) 108 / 112 / 113 (1)	ACCEPT 4.8 cm converted to 48 000 μm ecf for measurement of 9.5 cm or 9.7 cm to max 2 marks ecf for using diameter not radius	(3)

Question number	Answer	Additional guidance	Mark
4(a)(iv)	An explanation that includes the following points: - so that {more light <u>absorbed</u> / more chlorophylls are present} (1) - to generate {lots of / more} ATP / reduced NADP) / so that lots of glucose can be synthesised / more biomass produced} (1)	ACCEPT more for lots more {(photosynthetic) pigments / PSs / electron carrier proteins / ATP synthase} ACCEPT more for lots GALP for glucose	(2)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An answer that includes the following points: (two cycles requires) (6 + 12) 18 ATP and (6 + 6) 12 NADPH (1) 1.5 : 1 (1)	ACCEPT 9 : 6 etc but not 6 : 9 etc values shown for one cycle or more	(2)

Question number	Answer	Mark
4(b)(i)	The only correct answer is D <i>A is incorrect because ADP is not hydrolysed and energy is not released</i> <i>B is incorrect because ADP is not hydrolysed</i> <i>C is incorrect because energy is not released</i>	(1)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	An explanation that includes the following points: - because {it is used to regenerate RuBP / (otherwise) the cell will run out of RuBP} (1) (with no RuBP) no {carbon fixation / (Calvin) cycle / light independent stages} will occur (1)	IGNORE any numbers given ACCEPT used to make {glycerol / amino acid / nucleotides} stand alone ACCEPT description of carbon fixation (regeneration of RuBP) for next cycle of carbon fixation	(2)